

Original theta moments: explicit error bounds and finite rational witnesses

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This calculation continues the supplied *RH Counterfactual Workbench*. It keeps its original $g = 2\xi$, theta source, Fourier coordinate, unscaled mass, reciprocal-zero multiplicities, and monomial matrices. The accompanying workbench gives the full arithmetic packet and support comparison. Here the infinite theta integral receives explicit truncation bounds, and the negative-witness construction receives quantitative rational approximation and finite interval verification. The computational certificate uses these bounds to prove positivity of both original matrices of dimension 16. This finite result has the same matrix definitions as the workbench's exploratory calculation.

1 The original integral and its two omitted regions

Retain

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \psi(y) = e^{y/2} f_0(e^y). \quad (\text{TC1})$$

The original reflection and Mellin calculation give $\psi(-y) = \psi(y)$ and $g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy$. For every even integer $J \geq 0$, write $M_J = \int_{\mathbb{R}} y^J \psi(y) dy$. Substitution of (TC1), with its theta factor two and reflection factor two, gives

$$M_J = \int_0^\infty I_J(y) dy, \quad I_J(y) = 4y^J \sum_{n \geq 1} \left(4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2} \right) e^{-\pi n^2 e^{2y}}. \quad (\text{TC2})$$

At $y \geq 0$, each summand is nonnegative; it is strictly positive at $y > 0$, because $2\pi n^2 e^{2y} - 3 > 0$. This also proves positivity of every M_J once convergence is established below.

For an integer $N \geq 1$ and a real $L > 0$, retain exactly the terms $1 \leq n < N$ on $0 \leq y \leq L$. Denote their integral by $M_J^{N,L}$. Thus N is the first omitted integer. Set

$$r_N = e^{4/N - 2\pi N}, \quad r_1 = e^{4 - 2\pi}, \quad \lambda_J(L) = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{TC3})$$

Both r_N and r_1 lie strictly between zero and one.

Theorem 1 (Two explicit positive tails). *For $\lambda_J(L) > 0$ the full original moment satisfies*

$$0 \leq M_J - M_J^{N,L} \leq T_{J,N,L} + Y_{J,L}, \quad (\text{TC4})$$

where every factor is retained in

$$T_{J,N,L} = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - r_N}, \quad Y_{J,L} = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - r_1)\lambda_J(L)}. \quad (\text{TC5})$$

Proof. For $n = N + \ell$, $\ell \geq 0$, use $(N + \ell)^4 \leq N^4 e^{4\ell/N}$, which follows from $\log(1 + x) \leq x$ for $x \geq 0$, and $(N + \ell)^2 \geq N^2 + 2N\ell$. For $y \geq 0$ these imply

$$\sum_{n \geq N} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{N^4 e^{-\pi N^2 e^{2y}}}{1 - e^{4/N - 2\pi N e^{2y}}} \leq \frac{N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TC6})$$

The negative term in each parenthesis of (TC2) can be dropped for an upper bound. On $[0, L]$ one has $y^J \leq L^J$ and $e^{9y/2} \leq e^{9L/2}$. Integrating that bound over an interval of length L proves the first expression in (TC5).

For the second region, apply the first bound in (TC6) with $N = 1$, retaining its numerator's dependence on y and bounding its denominator below by $1 - r_1$. Consequently

$$0 \leq I_J(y) \leq \frac{16\pi^2}{1 - r_1} y^J e^{9y/2 - \pi e^{2y}} \quad (y \geq 0). \quad (\text{TC7})$$

Write $y = L + v$ with $v \geq 0$. The elementary inequalities $(L + v)^J \leq L^J e^{Jv/L}$ and $e^{2(L+v)} \geq e^{2L}(1 + 2v)$ yield

$$y^J e^{9y/2 - \pi e^{2y}} \leq L^J e^{9L/2 - \pi e^{2L}} e^{-\lambda_J(L)v}. \quad (\text{TC8})$$

Its integral for $v \geq 0$ is the numerator divided by $\lambda_J(L)$, proving the second bound. The two omitted regions are disjoint: the first has $0 \leq y \leq L$ and $n \geq N$; the second has $y > L$ and every $n \geq 1$. Every omitted summand is nonnegative. Tonelli's theorem therefore justifies all exchanges before convergence is known; the finite upper bounds then prove convergence and (TC4). $\square$

For every fixed J , the bounds tend to zero by first increasing L and then increasing N . Indeed (TC8) is dominated by the negative double exponential as $L \rightarrow \infty$, while for fixed L the factor $N^4 e^{-\pi N^2}$ tends to zero and $r_N \rightarrow 0$. The finite integrand

$$y^J e^{y/2} \sum_{n=1}^{N-1} (16x_n(y)^2 - 24x_n(y)) e^{-x_n(y)}, \quad x_n(y) = \pi n^2 e^{2y}, \quad (\text{TC9})$$

is entire as a function of the original complex coordinate y . Expanding x_n shows that it is exactly the retained integrand of (TC2); no scalar factor or negative term has been changed. This entire extension supplies the analyticity condition for validated complex-ball quadrature on the finite real intervals.

2 Exact propagation to the original Hankel matrices

Define, in the original coefficient convention,

$$a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad F(w) = \sum_{j \geq 0} a_j w^j, \quad -\frac{F'(w)}{F(w)} = \sum_{n \geq 0} b_n w^n. \quad (\text{TC10})$$

The original entire-function comparison is $F(z^2) = g(1/2 + iz)$, and $a_0 = M_0 > 0$. Multiplication of the last identity by F and extraction of the coefficient of w^n prove the complete recurrence

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{TC11})$$

In particular the original mass remains in every division by a_0 . The two real symmetric matrices of dimension $d + 1$ are

$$H_d = (b_{r+s})_{0 \leq r, s \leq d}, \quad H_d^{(1)} = (b_{r+s+1})_{0 \leq r, s \leq d}. \quad (\text{TC12})$$

Proposition 2 (Finite enclosures and signs). *Suppose that closed rational intervals enclose the finite integrals and upper bounds in (TC4)–(TC5). Adding the interval $[0, T_{J,N,L} + Y_{J,L}]$ with outward upper endpoints gives rational enclosures of the full moments. Applying (TC10)–(TC11) by interval arithmetic, with an enclosure of a_0 having positive lower endpoint, gives enclosures of every required b_n . An interval LDL^T recursion whose every pivot has positive lower endpoint proves positive definiteness of the original matrix in (TC12).*

Proof. Addition, subtraction, multiplication and division by an interval excluding zero each contain the exact operation on contained arguments. This follows for multiplication from the extrema of a bilinear function on a rectangle; division reduces to multiplication by the interval reciprocal. Induction in (TC11) proves the assertion. No independence of the interval arguments is required for enclosure.

For a symmetric matrix A , the exact recursion in the unchanged coordinate order is

$$\begin{aligned} D_j &= A_{jj} - \sum_{k < j} L_{jk}^2 D_k, \\ L_{jj} &= 1, \quad L_{ij} = \frac{A_{ij} - \sum_{k < j} L_{ik} L_{jk} D_k}{D_j} \quad (i > j), \quad L_{ij} = 0 \quad (i < j). \end{aligned} \quad (\text{TC13})$$

If the interval pivots have positive lower endpoints, induction shows that every exact D_j is positive and every division is defined. Formula (TC13) makes $A_{ij} = \sum_k L_{ik} D_k L_{jk}$ for all $i \geq j$; symmetry gives the remaining entries. Thus $A = LDL^T$. The triangular L is invertible with determinant one. For every nonzero real vector v , $v^T A v = \sum_j D_j ((L^T v)_j)^2 > 0$. The same equality for complex vectors uses the conjugate transpose and squared moduli and proves Hermitian positive definiteness. $\square$

For a specified rational vector $c = (c_0, \dots, c_d)$ let $\gamma_n = \sum_{r+s=n} c_r c_s \in \mathbb{Q}$. The exact signed witness is

$$c^T H_d c = \sum_{n=0}^{2d} \gamma_n b_n. \quad (\text{TC14})$$

If $b_n \in [\ell_n, u_n]$, its upper endpoint is

$$U(c) = \sum_{\gamma_n \geq 0} \gamma_n u_n + \sum_{\gamma_n < 0} \gamma_n \ell_n. \quad (\text{TC15})$$

Every comparison here is rational. A strict inequality $U(c) < 0$ is a finite negative certificate. If $\hat{b}_n$ have errors $|b_n - \hat{b}_n| \leq \epsilon_n$, the exact error in (TC14) is at most $\sum_n |\gamma_n| \epsilon_n$, by the triangle inequality. These statements include all mixed matrix entries.

3 Quantitative rational form of the counterfactual separator

We record the full argument to show precisely how a negative counterfactual form becomes a rational certificate. The entire factorization in the supplied workbench gives the distinct reciprocal zeros β_j of F , with positive integer multiplicities m_j , and

$$\sum_j m_j |\beta_j| < \infty, \quad b_n = \sum_j m_j \beta_j^{n+1}. \quad (\text{TC16})$$

No reciprocal zero is zero. The multiset is stable under complex conjugation. The actual map from a zeta zero $\rho = 1/2 + \delta + i\gamma$ is

$$\beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \quad (\text{TC17})$$

The reflected pair $\rho, 1 - \rho$ corresponds to a single zero of F with the same multiplicity. Both primary blocks remain in the original packet before taking its half-trace.

Let

$$S_* = \sum_j m_j |\beta_j|, \quad R_* = \max\{1, \sup_j |\beta_j|\} < \infty, \quad \|P\|_{R_*} = \sum_{r=0}^{\deg P} |p_r| R_*^r \quad (\text{TC18})$$

for $P(X) = \sum p_r X^r$. This coefficient norm is an explicit error-control map; it changes neither P nor the original Hankel form. Define

$$\mathcal{Q}(P) = \sum_j m_j \beta_j P(\beta_j)^2 = p^T H_{\deg P} p. \quad (\text{TC19})$$

Absolute convergence follows from $|P(\beta_j)| \leq \|P\|_{R_*}$ and (TC16). The sum is real for real P because conjugate terms pair, and real reciprocal terms are real. The equality with the finite matrix follows by expanding the finite polynomial product and using (TC16).

Lemma 3 (Coefficient perturbation with the full reciprocal tail). *For real polynomials P, E one has*

$$|\mathcal{Q}(P + E) - \mathcal{Q}(P)| \leq S_* \|E\|_{R_*} (2\|P\|_{R_*} + \|E\|_{R_*}). \quad (\text{TC20})$$

Proof. The polynomial identity $(P + E)^2 - P^2 = 2PE + E^2$ holds at every β_j including every multiplicity. Take absolute values term by term in the absolutely convergent series (TC19), bound $|P(\beta_j)|$ and $|E(\beta_j)|$ by (TC18), and sum $m_j |\beta_j|$. This gives (TC20). $\square$

Theorem 4 (Explicit finite degree and rational coefficient accuracy). *Choose any nonreal reciprocal zero β and its conjugate of multiplicity m . Put $r = |\beta|/2$. Let $\mathcal{A}$ be the finite set of all other distinct reciprocal zeros of modulus at least r , and set*

$$B(X) = \prod_{\alpha \in \mathcal{A}} (X - \alpha), \quad B_0 = \prod_{\alpha \in \mathcal{A}} (r + |\alpha|), \quad T_* = \sum_{|\beta_j| < r} m_j |\beta_j|. \quad (\text{TC21})$$

Choose either $c \in \mathbb{C}$ satisfying $\beta c^2 = -1$ and define

$$C_* = \frac{|c|B_0}{|B(\beta)|} \left(1 + \frac{|\Re \beta| + r}{|\Im \beta|} \right). \quad (\text{TC22})$$

If $T_* = 0$, take $N_* = 0$. Otherwise take the explicit integer

$$N_* = \max \left\{ 0, 1 + \left\lfloor \frac{\log(C_*^2 T_*/m)}{\log 4} \right\rfloor \right\}. \quad (\text{TC23})$$

For any integer $N \geq N_*$ put

$$w_N = \frac{c}{\beta^N B(\beta)}, \quad v_N = \frac{\Im w_N}{\Im \beta}, \quad u_N = \Re w_N - (\Re \beta) v_N, \quad P_N(X) = X^N B(X) (u_N + v_N X). \quad (\text{TC24})$$

Then $\mathcal{Q}(P_N) \leq -m$. A rational polynomial $\tilde{P}_N$ of degree at most $D = N + \#\mathcal{A} + 1$ can be obtained by rounding each of the $D + 1$ original coefficients to a nearest multiple of $1/H$, with either choice allowed at a tie, for any positive integer H with

$$\frac{\sum_{r'=0}^D R_*^{r'}}{2H} \leq \min \left\{ 1, \frac{m}{2S_*(2\|P_N\|_{R_*} + 1)} \right\}. \quad (\text{TC25})$$

Every such rounding satisfies $\mathcal{Q}(\tilde{P}_N) \leq -m/2 < 0$.

Proof. Absolute summability in (TC16) makes $\mathcal{A}$ finite. It is conjugation stable; hence B has real coefficients, $B(\beta) \neq 0$, and u_N, v_N are real. Evaluation of their formulas gives $u_N + v_N \beta = w_N$ and therefore $P_N(\beta) = c$ and $P_N(\bar{\beta}) = \bar{c}$. The selected two terms of (TC19) sum to $-2m$ exactly. All other terms with modulus at least r vanish.

For $|X| \leq r$, one has $|B(X)| \leq B_0$ and

$$|u_N + v_N X| \leq |w_N| \left(1 + \frac{|\Re \beta| + r}{|\Im \beta|} \right).$$

Consequently $|P_N(X)| \leq C_* 2^{-N}$. The absolute sum of all remaining terms is at most $C_*^2 T_* 4^{-N}$, so

$$\mathcal{Q}(P_N) \leq -2m + C_*^2 T_* 4^{-N}. \quad (\text{TC26})$$

The chosen integer (TC23) makes the last summand at most m ; if $T_* = 0$ it is zero. This proves the first assertion. The coefficient rounding error $E = \tilde{P}_N - P_N$ has each coefficient at most $1/(2H)$ in absolute value. Thus $\|E\|_{R_*} \leq \sum_{r'=0}^D R_*^{r'}/(2H)$. The bound in (TC25) and (TC20) make the change in the quadratic form at most $m/2$, proving the strict negative result. Every original coefficient slot is included in the error bound. A nonzero coefficient may round to zero; an original zero coefficient remains an exact zero. $\square$

All constants in this counterfactual degree bound retain their dependence on the chosen actual zero and on the full reciprocal set. It supplies no universal dimension for an off-critical witness. Once a rational polynomial is specified, however, (TC14)–(TC15) require only finitely many original moments. As their enclosing intervals shrink, their propagated intervals shrink to the exact coefficients: (TC11) is a finite composition of continuous rational operations with denominator $a_0 > 0$. The strict margin in (TC25) therefore eventually yields a strictly negative rational upper bound in (TC15).

4 The completed finite certificate

The supplied computation uses $N = 20$, $L = 4$, and $J = 0, 2, \dots, 64$. It keeps all nineteen original theta terms on each of the eight intervals $[j/2, (j+1)/2]$, $0 \leq j < 8$. The omitted two regions are enclosed by (TC5). For these parameters every $\lambda_J(L)$ is positive: already $\pi > 3$ and $e^8 > 1 + 8 + 8^2/2 = 41$ give $2\pi e^8 - 9/2 - J/4 > 246 - 9/2 - 16 > 0$. The actual computation retains ball enclosures of the complete expressions, rather than replacing them by that rough inequality.

The finite integrations use python-flint 0.9.0's validated `acb.integral`, at 1024 bits and target 850 bits. The callback is exactly (TC9); its entire expression satisfies the integration routine's analytic-ball contract. Each returned real interval is enlarged by the positive tails. All endpoints are exported as exact dyadic rational numerators and denominators. The complete a_j, b_n, L_{ij}, D_j enclosures and the individual integration intervals are retained in the machine-readable certificate. All thirty-two computed pivots have strictly positive lower endpoints, proving

$$H_{15} > 0, \quad H_{15}^{(1)} > 0. \quad (\text{TC27})$$

Every smaller leading principal matrix inherits positivity. For orientation, the readable enclosures include

$$\det H_1 \in \left[1.9493355548191422147398782191721395 \times 10^{-9} \pm 5.60 \times 10^{-45} \right]. \quad (\text{TC28})$$

The actual certificate contains the more precise exact rational endpoints. Its last pivots are positive intervals centered at $6.1698420811114408502314906319786538 \times 10^{-107}$ and $7.1026100619440799659087832255229554 \times 10^{-111}$, respectively. The original a_0 remains centered at 0.99424155637662821982554747937079544. These values are given to identify the checked objects; the strict signs are decided by the enclosing rational endpoints.

The complete workbench proves that positivity for every unshifted dimension is equivalent to RH. Equation (TC27) settles the displayed finite dimensions and makes no assertion about larger dimensions. The packet trace still evaluates the retained arithmetic source before any critical-only comparison, and its support maps retain τ and every receiving supported zero exactly as in the included packet proofs.

References

- [1] *RH Counterfactual Workbench*, original supplied package, 13 September 2026, complete note and calculation sources.
- [2] The python-flint developers, *acb: complex ball arithmetic and validated integration*, <https://python-flint.readthedocs.io/en/latest/acb.html>.
- [3] R. Zhang, *On Power Sums of Positive Numbers*, <https://arxiv.org/abs/1510.03420>.